

TABLE III
Wireless Charging Solutions for Design Engineers

Manufacturer	Solution	Feature	Advantages
Freescale	MWCT1000CFM	5W single-coil transmitter	• Robust foreign object detection algorithm to meet latest safety requirements • Transfer efficiency exceeding 75%
NXP	NXQ1TXA5	Qi low-power A5/A11 transmitter according to specification version 1.1.2	• Extremely small PCB size • Safe operation with FOD and optional NTC
Active-Semi	EVK-PAC5220WP-Qi-xxA11-V1	Single-4C based wireless power transmitter	• Built-in surge protection • Scalable MCU based solution to enable products with multi-coils, multi-modes, and medium or higher power levels up to 150W
NXP	NXQ1TXA6	12V multi-coil wireless charging transmitter	• Low standby power and NFC-enabled zero-power standby • Optional smart NFC features like Bluetooth pairing
Active-Semi	PAC5220WP	Power application controller	• 28 GPIOs for MCUs • Interface type: SPI, I2C, UART, SWD
PowerbyProxi	Proxi-Ring 480W	A wireless slip ring	• Transfer power through any non-metallic material including liquids, solids and gases • Dual channels provide 480W with no redundancy or 240W with dual redundancy
PowerbyProxi	Proxi-Point	A wireless connector	• Reduces high-installation and on-going maintenance costs • Simplifies implementation/installation processes
Texas Instruments	bq51020EVM-520	Wireless power receiver evaluation module	• Inductor-less receiver for lowest height profile solution • Adjustable output voltage (4.5V to 8V) for coil and thermal optimisation
Texas Instruments	BQ51221	Dual-mode, 5W (WPC and PMA) single-chip wireless power receiver	• Fully synchronous rectifier with 96% efficiency • 79% system efficiency at 5W
STMicroelectronics	STEVAL-IDS002V1	Autonomous wireless multi-sensor node powered by PV cells	• User-friendly software GUI for system configuration • PV module soldered on top
Broadcom	BCM59350	A wireless charging PMU	• Up to 88% AC-to-DC efficiency • Ensures acceptable NFC performance

based on Qi and Power Matter Alliance (PMA) standards are already in market, the products based on A4WP are yet to be seen.

Verizon and DOCOMO are using Qi standard, AT&T has chosen PMA standard and T-Mobile wants to use A4WP. Power Matters brings various PMA-based wireless charging options like Power snap kit, travel mat, power mat and access kit for high-end mobile devices.

Anand says, "While there are adequate tools and resources available from various chip manufacturers for designing the wireless charging devices, on-going battle and unstable standards are restrictive. It is difficult for the designers to neglect one or two particular standards that in turn would add to the cost of product and design activity. Having said that, it is still less of a design problem and more of an innovative utilisation problem for the

Share Your Knowledge

If you are an expert in your field of electronics, why not share your knowledge with the world?



Electronics For You invites electronics professionals and industry experts to write articles on their area of expertise and interest from different perspectives

Some of the topics we want you to write about are:

- Photonics: Optics & Optoelectronics
- Automation
- Power Electronics
- Components
- Telecom



So if you have an interesting topic or original article to share, please let us know at efyeditorial@efy.in

We would love to hear your good ideas too!

electronics
FOR YOU